

CHOIR, Mongolia

WMO: 442980

Lat: 46.3500N

Lon: 108.3747E

Elev: 4219

StdP: 12.59

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-18.0	-26.8	1.4	-19.2	-23.2	1.8	-15.9	26.9	-1.1	22.9	-2.6	10.0	0	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	88.8	62.7	85.2	61.3	81.8	60.1	66.5	81.6	64.3	78.7	62.5	76.6	9.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	96.1	72.6	59.4	88.4	70.5	57.3	81.9	68.9	33.7	81.5	31.8	79.1	30.4	76.8	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	22.9	20.7	-26.1	93.6	4.9	4.0	-29.6	96.5	-32.5	98.8	-35.2	101.1	-38.7	104.0
			-26.3	70.7	4.8	4.8	-29.8	74.2	-32.6	77.0	-35.3	79.7	-38.8	83.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	-3.2	5.3	22.6	39.9	52.7	64.4	69.8	65.6	54.2	37.0	15.6	0.8
	DBStd	27.02	8.43	10.42	11.22	9.64	9.24	7.44	6.15	6.73	8.70	9.42	11.75	8.61
	HDD50	7176	1649	1252	848	323	78	2	0	0	56	411	1031	1526
	HDD65	11112	2114	1672	1313	753	393	100	20	74	333	868	1481	1991
	CDD50	1903	0	0	0	20	162	432	613	485	183	8	0	0
	CDD65	366	0	0	0	0	12	81	169	94	10	0	0	0
	CDH74	3574	0	0	0	3	210	864	1579	813	105	0	0	0
	CDH80	1121	0	0	0	0	46	269	562	232	12	0	0	0
Wind	WSAvg	8.2	7.5	7.5	8.7	10.4	10.7	9.2	7.8	7.1	8.0	7.8	6.9	7.0
Precipitation	PrecAvg	5.8	0.0	0.0	0.1	0.2	0.4	1.0	1.9	1.2	0.6	0.2	0.1	0.1
	PrecMax	11.3	0.2	0.3	0.6	0.5	1.3	2.0	4.8	3.2	3.1	0.5	1.3	0.2
	PrecMin	2.8	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.1	0.0	0.0	0.0	0.0
	PrecStd	2.0	0.0	0.1	0.1	0.2	0.4	0.6	1.2	0.8	0.6	0.2	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.4	35.6	58.2	72.1	85.1	91.6	94.5	91.0	81.6	67.2	46.7	25.3
		MCWB	18.6	28.0	41.0	50.3	56.5	63.1	67.5	63.7	57.8	48.8	36.2	21.4
	2%	DB	17.3	30.1	50.0	66.2	79.9	86.6	90.0	85.7	77.1	60.7	41.4	20.4
		MCWB	14.2	24.1	36.6	47.0	54.9	59.6	64.3	62.1	56.0	45.4	33.0	17.2
	5%	DB	12.9	25.0	44.7	61.4	75.2	82.7	86.2	81.8	72.7	56.0	36.5	16.7
		MCWB	10.5	20.8	33.2	43.8	52.4	59.3	63.0	61.2	53.7	42.5	29.8	14.2
	10%	DB	9.0	20.7	39.9	56.9	70.4	78.9	82.4	78.3	68.7	51.7	32.1	13.2
		MCWB	7.2	17.7	30.5	41.7	50.7	57.9	61.5	59.7	52.0	40.0	26.6	11.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.2	28.6	41.7	52.2	59.5	67.4	71.9	69.0	61.1	50.2	36.8	21.9
		MCDB	22.9	35.3	56.8	70.0	80.4	83.6	88.8	83.6	75.1	65.5	45.0	25.2
	2%	WB	14.4	24.3	37.2	47.9	56.3	63.3	67.4	65.6	58.0	46.3	33.2	17.5
		MCDB	17.1	29.7	49.4	64.4	76.9	79.7	82.8	80.1	72.7	59.1	40.9	20.2
	5%	WB	10.6	21.0	33.7	44.8	53.7	61.3	65.1	63.3	55.5	43.2	29.9	14.2
		MCDB	12.8	24.7	44.4	60.5	72.4	77.5	79.7	76.6	69.5	54.9	36.0	16.6
	10%	WB	7.2	17.6	30.6	42.0	51.2	59.6	63.3	61.3	53.0	40.4	26.9	11.2
		MCDB	8.9	20.7	39.4	56.0	68.8	75.2	77.5	74.0	66.7	50.8	32.0	13.1
Mean Daily Temperature Range	5% DB	MDBR	15.7	19.9	22.5	23.1	23.7	21.3	19.5	19.4	21.2	20.7	18.0	14.7
		MCDBR	19.4	23.7	26.9	28.8	29.0	25.7	23.6	23.2	25.4	25.1	21.8	17.5
		MCWBR	17.2	19.3	17.9	17.3	15.8	11.8	10.5	11.1	13.1	15.3	15.6	15.3
	5% WB	MCDBR	19.1	23.0	26.3	27.7	27.3	22.0	20.7	20.2	22.6	24.1	21.2	17.5
		MCWBR	17.1	18.9	18.0	17.5	16.2	11.6	10.8	10.9	12.8	15.5	15.6	15.3
Clear-Sky Solar Irradiance	taub	0.241	0.265	0.303	0.351	0.390	0.407	0.381	0.363	0.323	0.287	0.256	0.241	
	taud	2.376	2.328	2.279	2.193	2.119	2.173	2.307	2.332	2.383	2.427	2.406	2.369	
	Ebn at Noon	279	292	293	286	277	272	279	279	283	280	270	266	
	Edh at Noon	22	29	36	43	48	46	39	37	32	26	21	20	
All-Sky Solar Radiation	RadAvg	637	983	1480	1907	2192	2172	2069	1875	1562	1118	717	524	
	RadStd	56	77	68	79	111	115	94	82	74	49	40	44	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-2.18	N/A	N/A	N/A	N/A	
(47)	Regional (2 neighbors)	N/A	N/A	-3.46	N/A	-1.80	-1.64	N/A	N/A	N/A	N/A	

Nomenclature: See separate page